Installing KVM on Ubuntu Server

Complete Step-by-Step Installation Guide

Compatible with Ubuntu 20.04 / 22.04 / 24.04 LTS

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What is KVM?

KVM (Kernel-based Virtual Machine) is an open-source virtualization technology built directly into the Linux kernel. It turns your Ubuntu Server into a full hypervisor, allowing you to run multiple isolated virtual machines (VMs) on a single physical machine.

KVM works together with:

- QEMU — the hardware emulator that runs the VMs
- libvirt — a management layer that provides tools like virsh and virt-install
- bridge-utils — for configuring VM networking

> ⚠ **WARNING**
> KVM requires hardware virtualization support (Intel VT-x or AMD-V). This must be enabled in your BIOS/UEFI before proceeding.

Prerequisites

#	Requirement	Details
1	**Ubuntu Server**	20.04, 22.04, or 24.04 LTS (64-bit)
2	**CPU with VT-x or AMD-V**	Check in BIOS — must be enabled
3	**Minimum 4 GB RAM**	2 GB for host + 2 GB for each VM
4	**Minimum 20 GB disk**	For host OS + VM disk images
5	**sudo / root access**	All commands use sudo
6	**Internet connection**	To download packages from apt

Step 1 — Check CPU Virtualization Support

Before installing anything, verify that your CPU supports hardware virtualization:

```
grep -c vmx /proc/cpuinfo    # Intel VT-x
grep -c svm /proc/cpuinfo    # AMD-V
```

If the output is 0 for both, go into your BIOS/UEFI settings and enable Intel Virtualization Technology (VT-x) or AMD-V, then reboot and try again.

Install and run the official KVM checker:

```
sudo apt install -y cpu-checker
```

```
sudo kvm-ok
```

> ✓ **SUCCESS**
> Expected output: INFO: /dev/kvm exists — KVM acceleration can be used If you see 'KVM acceleration can NOT be used', fix your BIOS settings first.

Step 2 — Update the System

Always update your package list and system before installing new software:

```
sudo apt update
sudo apt upgrade -y
```

Step 3 — Install KVM and All Required Packages

Install everything in one command:

```
sudo apt install -y \
  qemu-kvm \
  libvirt-daemon-system \
  libvirt-clients \
  bridge-utils \
  virt-install \
  virt-viewer \
  virtinst
```

Here is what each package does:

Package	Purpose
`qemu-kvm`	The core QEMU/KVM hypervisor — runs the VMs
`libvirt-daemon-system`	The libvirt daemon (libvirtd) — manages VMs as a service
`libvirt-clients`	Command-line tools including virsh
`bridge-utils`	Tools for creating and managing network bridges (brctl)
`virt-install`	Command-line tool to create new VMs
`virt-viewer`	GUI console to connect to a running VM
`virtinst`	Additional Python library used by virt-install

Step 4 — Enable and Start the libvirt Service

The libvirtd service must be running for KVM to work. Enable it to start automatically on boot:

```
sudo systemctl enable libvirtd
sudo systemctl start libvirtd

# Verify it is running:
sudo systemctl status libvirtd
```

> ✓ **SUCCESS**
> Expected output: Active: active (running). If it shows failed, check logs with: sudo journalctl -u libvirtd -n 50

Step 5 — Add Your User to the Required Groups

By default, only root can manage VMs. Add your user to the libvirt and kvm groups so you can use virsh and virt-install without sudo every time:

```
sudo adduser $USER libvirt
sudo adduser $USER kvm

# Apply the new group membership without logging out:
newgrp libvirt
```

> ℹ **NOTE**
> To make this permanent, log out and log back in after running the adduser commands.

Step 6 — Verify the Installation

Run these checks one by one to confirm KVM is fully installed and working:

6.1 Check KVM kernel modules are loaded

```
lsmod | grep kvm
# Expected output:
# kvm_intel    xxx    0       (or kvm_amd for AMD CPUs)
# kvm          xxx    1   kvm_intel
```

6.2 Check /dev/kvm device exists

```
ls -la /dev/kvm
# Expected: crw-rw---- 1 root kvm  ...
```

6.3 Verify libvirt is working

```
virsh list --all
# Expected: empty table (no VMs yet) — but no errors

# Also check libvirt version:
virsh version
```

6.4 Check installed package versions

```
dpkg -l | grep -E 'qemu-kvm|libvirt-daemon|virt-install'
# All lines should start with 'ii' (installed)
```

6.5 Check default network

```
sudo virsh net-list --all
# Expected:
# Name      State    Autostart   Persistent
# default   active   yes         yes
```

> **ℹ NOTE**
> The 'default' network (NAT, 192.168.122.x) is created automatically by libvirt. If it is not active, run: sudo virsh net-start default && sudo virsh net-autostart default

Step 7 — Quick Test: Create a Minimal VM

To confirm everything works end-to-end, create a small test VM. You will need an ISO file — for example Ubuntu 22.04:

```
# Download Ubuntu 22.04 minimal ISO (optional — use any ISO you have)
wget https://releases.ubuntu.com/22.04/ubuntu-22.04.5-live-server-amd64.iso \
  -P /home/$USER/
```

Create the test VM:

```
sudo virt-install \
```

```
  --name test-vm \
  --memory 1024 \
  --vcpus 1 \
  --disk path=/var/lib/libvirt/images/test-vm.qcow2,size=10 \
  --os-variant ubuntu22.04 \
  --cdrom /home/$USER/ubuntu-22.04.5-live-server-amd64.iso \
  --network network=default \
  --graphics vnc \
  --noautoconsole
```

Verify the VM was created and is running:

```
virsh list --all
# Expected:
# Id   Name      State
# 1    test-vm   running
```

Connect to the VM console to see it boot:

```
virt-viewer --connect qemu:///system test-vm
```

When done testing, remove the test VM:

```
sudo virsh destroy test-vm
sudo virsh undefine test-vm --remove-all-storage
```

> ✓ **SUCCESS**
> KVM is fully installed and working if the test VM boots successfully.

Troubleshooting

Problem	Fix
kvm-ok returns 'cannot use KVM'	Enable VT-x or AMD-V in BIOS/UEFI settings, then reboot
/dev/kvm does not exist	Run: sudo modprobe kvm kvm_intel (or kvm_amd for AMD)
libvirtd failed to start	Run: sudo journalctl -u libvirtd -n 50 to see the error
virsh: Permission denied	Make sure your user is in the libvirt group, then log out and back in
virt-install: --os-variant error	Run: sudo apt install -y osinfo-db then retry

default network is inactive	Run: sudo virsh net-start default && sudo virsh net-autostart default
VM disk image: Permission denied	Run: sudo chmod 755 /var/lib/libvirt/images/

Quick Reference — Essential Commands After Installation

Command	Purpose
`sudo kvm-ok`	Confirm KVM is usable
`lsmod | grep kvm`	Check KVM kernel modules
`sudo systemctl status libvirtd`	Check libvirt service status
`virsh list --all`	List all VMs
`sudo virsh net-list --all`	List all virtual networks
`virsh version`	Show KVM/libvirt version
`virt-install --os-variant list`	List all supported OS variants
`sudo virsh destroy <vm>`	Force stop a VM
`sudo virsh undefine <vm> --remove-all-storage`	Delete a VM and its disk

Conclusion

After completing this guide you have:

- Verified your CPU supports hardware virtualization
- Installed QEMU-KVM, libvirt, virt-install and bridge-utils
- Enabled the libvirtd service to start on boot
- Added your user to the libvirt and kvm groups
- Verified the installation with virsh and a test VM

Your Ubuntu Server is now a fully functional KVM hypervisor. You can proceed to create virtual machines and configure bridged networking to connect them to your LAN.